MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION (COMBINED TECHNICAL) FOR RECRUITMENT TO THE POSTS OF AGRICULTURE & ALLIED SERVICES, GOVERNMENT OF MIZORAM.

(NOVEMBER-2025)

AGRICULTURE SCIENCE PAPER-III

Time Allowed : 3 hours FM : 200

SECTION - A *(Multiple Choice questions) (100 Marks)*

All questions carry equal mark of 2 each. Attempt all questions.

*This Section should be answered only on the **OMR Response Sheet** provided.*

Notes regarding Negative Marking :

(i) There will be penalty (negative marking) for every wrong answer given by a candidate in the MCQ type questions.

(ii) For each question for which a wrong answer has been given by the candidate, one-third of the marks assigned to that question will be deducted as penalty

(iii) If a candidate gives more than one answer to one question, it will be treated as a wrong answer even if any one of the given answers happens to be correct and there will be same penalty as above for that question.

(iv) If a question is left blank, i.e., no answer is given by the candidate, there will be no penalty for that question.

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1. _____ is the systematic process of collecting, analyzing and using information to track a programme's progress toward reaching its objectives and to guide management decisions.

(a) Monitoring (b) Planning

(c) Controlling (d) Directing

2. Use of physical space in communication process is called:

(a) Chronemics (b) Haptics

(c) Proxemics (d) Kinesics

3. The economic principle of _____ solves the problem of 'how to produce'.

(a) equi-marginal returns (b) factor substitution

(c) product substitution (d) returns to scale

4. HACCP defined as codex.

(a) Identifies and evaluates the hazards

(b) Identifies and controls hazards

(c) Identifies, evaluates and controls hazards

(d) Identifies and controls hazards after verification

5. The inanimate source of power utilized for doing agricultural operation is:

(a) Anima power (b) Human power

(c) Machine power (d) Wind power

6. Which model of communication is called the 'Mathematical Theory'?

(a) Shannon and Weaver (b) Berlo

(c) Schramm (d) Leagans

7. The energy spent in applying 100 Newton force to a body with its linear displacement of 100 meters is:

(a) 1000 Joule (b) 10000 Joule

(c) 5000 Joule (d) 100 Joule

8. APEDA replaced:

(a) FPO (b) FSSAI

(c) PFEPC (d) FAA

9. Mark the undiscounted technique to calculate feasibility of the investment.

(a) Net Present Value (b) Internal Rate of Return

(c) Pay Back Period (d) Benefit Cost Ratio

10. The concept of 'transactional model of communication' was introduced by:

(a) Shannon and Weaver (b) Wilbur Schramm

(c) Alder and Towne (d) Barnlund

11. Improved sickle used in hills is:

(a) Serrated sickle (b) Smooth sickle

(c) Rotary sickle (d) None of the above

12. Valorization of waste is a:

(a) Process of reusing waste material

(b) Process of recycling and composting waste material

(c) Process of reusing and composting waste material

(d) Process of reusing, recycling and composting waste material

13. Which describes the characteristics of 'Early Majority' the best?

(a) Traditional (b) Skeptical

(c) Deliberate (d) Venturesome

14. The additional cost associated with the use of an additional unit of input is:

(a) Marginal product cost (b) Marginal revenue

(c) Average cost (d) Marginal factor cost

15. Marketing mix is defined in terms of the following:

(a) 3 C's (b) 7 C's

(c) 4 P's (d) 3 R's

16. The study of individual organisms in relation to their environment is called:

(a) Synecology (b) Autecology

(c) Habitat ecology (d) Ecosystem ecology

17. How many bio-geographical zones are in India?

(a) 10 (b) 20

(c) 25 (d) 35

18. Which of the term is used for implement/ tools efficiency?

(a) Field capacity (b) Hp

(c) Cut off (d) Power

19. The main advantage of Farmer Field School (FFS) approach in agriculture extension is:

(a) Low cost (b) High participation

(c) Rapid dissemination (d) Individual focus

20. The most harmful ultraviolet radiations are:

(a) UV-C (b) UV-B

(c) UV-A (d) All of the above

21. Precursor of ethylene is:

(a) Tryptophane (b) Methionine

(c) ABA (d) IAA

22. The difference between the price paid and received by a specific marketing agency is:

(a) Price spread (b) Market margin

(c) Marketing cost (d) Consumer surplus

23. In an ecotone, the species which become abundant are called:

(a) Edge species (b) Keystone species

(c) Endemic species (d) Foster species

24. The most commonly used method for needs assessment in rural communities is:

(a) Random sampling (b) PRA

(c) Case studies (d) Focus group discussion

25. Which seeding method uses *Malobansa*?

(a) Transplanting (b) Seed dropping behind the plough

(c) Check row planting (d) Hill dropping

26. A visual which contains graphically certain process, like the life cycle of pests and pest control techniques is:

(a) Bulletin board (b) Handouts

(c) Wall chart (d) Mobile units

27. International Day for the Prevention of the Ozone Layer is observed on:

(a) 3rd January (b) 16th September

(c) 10th November (d) 26th March

28. ISO stands for:

(a) International Standard of Organization
(b) Indian Standard of Organization
(c) International Organization for Standard
(d) Indian Organization of Standard

29. In tractor engine, the complete path of power from the engine to the wheel is called:

(a) Power train
(b) Power wheel
(c) Both (a) and (b)
(d) None of the above

30. Rotavator for hill agriculture is usually:

(a) Tractor mounted
(b) Power tiller mounted
(c) Manual
(d) Animal drawn

31. The 'ceteris paribus' clause in the law of demand does not mean that:

(a) The taste of the consumer does not change
(b) The income of the consumer does not change
(c) The price of the consumer does not change
(d) The price of the substitute does not change

32. Dairy farming, poultry farming, crop farming and pig farming are:

(a) Systems of farming
(b) Types of farming
(c) Both (a) and (b)
(d) None of the above

33. Which theory emphasizes that adults are motivated to learn when they perceive the relevance of the content?

(a) Behaviourism
(b) Constructivism
(c) Andragogy
(d) Cognitivism

34. An AC motor is an electric motor powered by _____ type of current.

(a) Direct current
(b) Alternating current
(c) Pulsating current
(d) Continuous current

35. The most commonly used for grain storage to prevent pest infestation is:

(a) open sun drying
(b) Cold storage
(c) Hermetic storage
(d) Hydroponics

36. What is the temperature range typically used in pasteurization?

(a) 100-1200°C
(b) 60-750°C
(c) -18 to -230°C
(d) 200-2200°C

37. The Chipko movement is started by:

(a) Engler
(b) Sunderlal Bahuguna
(c) Medha Padkar
(d) None of these

38. Bulk density of food grains is defined as:

(a) Mass per unit volume including pores
(b) Mass per unit volume excluding pores
(c) Weight of 1000 kernels
(d) Porosity

39. The cost incurred by the farmer for cultivation of a crop per unit of output is:

(a) Yield
(b) Cost of production
(c) Cost of cultivation
(d) Total cost

40. _____ is the locus of points of different isoquants at which MRS of factors is constant.

(a) Iso-cost line
(b) PPC
(c) Expansion path
(d) Isoclines

41. The food chain in which micro-organisms breakdown dead producers is called:

(a) Consumer food chain
(b) Predator food chain
(c) Parasitic food chain
(d) Detritus food chain

42. Wheel hoe type weeder works on _____ of the operator.

(a) rotary action
(b) push and pull action
(c) sliding action
(d) vibratory action

43. Which storage method is commonly used to prevent rancidity in stored oils?

(a) Freezing
(b) Dry storage
(c) Refrigeration
(d) Hot storage

44. The name of the component placed in between driver wheels and differential of a tractor is:

(a) Gearbox
(b) Final drive
(c) Clutch
(d) Transmission shaft

45. The efficacy of HACCP system relies on:

(a) Management
(b) Employee
(c) Management and employee
(d) Lead auditor

46. Which leadership style is most effective in agricultural extension?

(a) Autocratic
(b) Democratic
(c) Laissez-faire
(d) Transactional

47. Minimum Support Price (MSP) is announced on the basis of the recommendation given by the:

(a) Ministry of Agriculture
(b) Chamber of Commerce
(c) Commission of Agricultural Costs and Prices
(d) Food and Agriculture Organization

48. Tendency of pollutants to become concentrated in successive trophic levels is known as:

(a) Biomagnification
(b) Biorythm
(c) Bioremediation
(d) Biopiracy

49. The main source of acid rain is:

(a) Sulphur dioxide
(b) Nitrogen
(c) Carbon dioxide
(d) Carbon monoxide

50. Who introduced the term Hot-spots as biodiversity?

(a) Charles Christopher Adam
(b) Norman Myers
(c) Aziz Absaber
(d) Warder Clyde Allee

SECTION - B *(Conventional Type) (100 Marks)*

*This Section should be answered only on the **Answer Sheet** provided.*
Marks for each question is indicated against it.
*Attempt **five (5)** questions.*

1. (a) Justify the statement 'Communication is a two way and dynamic process'. Elaborate any three (3) models of communication. **(3+3=6)**

(b) Defined farm management. Explain various types of farm management decisions. **(1+3=4)**

(c) Evaluate the impact of transgenic crops and animals on biodiversity and the environment. **(4)**

(d) Compare chemical and biological preservation methods used in food processing. **(4)**

(e) What are the challenges in farm mechanization in hilly and tribal areas? **(2)**

2. (a) What is the importance of PRA in extension? Explain three commonly used PRA tools. **(2+4=6)**

(b) Discuss the concept of crop insurance and its importance in mitigating risks in Indian agriculture. **(2+2=4)**

(c) What is greenhouse effect and how is it related to global warming? **(1+3=4)**

(d) What is the purpose of cushioning materials? How do you work out the static area load? **(2+2=4)**

(e) What is the role of power tillers in hill agriculture? **(2)**

3. (a) What is extension education? Discuss the contribution of agricultural extension research in agricultural development. **(2+4=6)**

(b) Define production function. Explain the three regions of production with the help of a graph. **(1+3=4)**

(c) What is carbon sequestration? Discuss its role in mitigating climate change. **(1+3=4)**

(d) Explain the various pre-harvest factors which affect the post-harvest life of a produce. **(4)**

(e) Write down the benefits of farm mechanization. **(2)**

4. (a) What are the perceived attributes of innovation? Classify the members of a social system on the basis of innovativeness using adopter category curve. **(2+4=6)**

(b) Explain the concept of Time Value of Money. Differentiate between compounding and discounting. **(2+2=4)**

(c) Define Hot-Spots of Biodiversity. Explain their significance in India. **(1+3=4)**

(d) What is precooling? Explain the different types of precooling. **(1+3=4)**

(e) Differentiate between primary and secondary tillage. **(2)**

5. (a) Define teaching and learning. What are the principles of extension teaching? **(2+4=6)**

(b) What do you understand by NABARD? Discuss in detail the role of NABARD in agriculture. **(1+3=4)**

(c) Define the term Biodiversity. What are the various threats to biodiversity? **(1+3=4)**

(d) Explain the importance of food safety standards in ensuring the safety of processed agricultural products. **(4)**

(e) Explain the significance of seed bed preparation. **(2)**

6. (a) Highlight the challenges faced by rural women in agriculture and the role of extension services in addressing them. **(6)**

(b) What is a farming system? Explain the concept and components of different farming systems with examples. **(1+3=4)**

(c) Why is 'Environmental Impact Assessment (EIA)' important? **(4)**

(d) Explain cold storage. Write down the basic principle of storage. **(1+3=4)**

(e) Describe the types of weeders suitable for small and marginal farmers. **(2)**

7. (a) What is capacity building in the context of extension? How does it improve agricultural productivity? **(2+4=6)**

(b) Discuss the types and importance of farm planning and budgeting in efficient farm management. **(4)**

(c) Examine the role of agriculture in both contributing to and combating climate change. **(4)**

(d) Discuss the principles of hygienic design in a food processing plant layout. **(4)**

(e) Differentiate between disc plough and disc harrow. **(2)**

8. (a) Explain extension teaching methods according to its form. Explain Audio visual aids as an extension teaching methods. **(3+3=6)**

(b) Note down the impact of World Trade Organization (WTO) on Indian agriculture. **(4)**

(c) Write down the effect of modern agricultural practices on soil pollution. **(4)**

(d) Discuss the role of post-harvest technology in doubling farmers' income. **(4)**

(e) Differentiate between indicated horse power and frictional horse power in I.C. **(2)**

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